INDUSTRY INTERNSHIP PRESENTATION

KAFILA: SMART FLEET MANAGEMENT SYSTEM

Enterprise-Grade iOS Platform for Logistics & Fleet Telematics

Presenter Information:

• Name: Ishan Kapoor (B.Tech Computer Science & Engineering)

• Institution: School of Computing Science & Engineering, Galgotias University

• Core Roles: Backend Integration, 2-Factor Authentication (2FA), & UI Testing

Host Organization:

• Infosys Limited — Global Education Centre (ETA), Mysore, Karnataka

Summer Internship Session 2025-2026 | Developed by 10-Member Agile Squad

ORGANIZATION

Infosys Limited — Corporate Overview & Mysore Campus

Global Technology Leader

- Founded in 1981 by N.R. Narayana Murthy.
- Over 315,000+ employees globally.
- Present across 56+ nations.
- Dual-listed on NYSE & NSE (INFY).
- $75B+ Enterprise Market Valuation.
- Clients include 1,800+ global enterprises.
- Pioneer of Global Delivery Model (GDM).

Mysore Global Campus (ETA)

- World's largest corporate academy.
- Spans 337 acres of world-class facilities.
- Trained 400,000+ IT engineers to date.
- High-intensity technical training pedagogy.
- Focus on full-stack, cloud & mobile stacks.
- State-of-the-art incubation labs.
- Agile Scrum project simulations.

Enterprise Focus Areas

- Digital Experience (Native iOS/Android).
- Infosys Cobalt (Cloud Enterprise Suite).
- Infosys Topaz (Generative AI platform).
- Cyber Security & Identity Governance.
- Quality Engineering & Test Automation.
- Real-time Logistics & Supply Chain IoT.
- Scalable Backend Microservices.

Project Context, Team Dynamics & Agile Execution

Internship Profile & Environment

- Role: iOS Development Intern.
- Division: Education, Training & Assessment (ETA), Mysore.
- Mentorship: Guided by Principal Solution Architects.
- Duration: 8 Weeks (June 2025 - August 2025).
- Working Model: Dedicated on-campus development labs.

10-Member Engineering Squad

- **10 Engineers cross-functionally organized.**
- **1 Lead Architect + 3 Feature Track Leads.**
- **3 Frontend UI Specialists (SwiftUI View Hierarchy).**
- **3 Backend, Security & Testing Leads (Ishan's Team).**
- **Git branching strategy: main, dev, feature/exp.**

Agile Scrum Delivery in Atlassian JIRA

- **Sprints: Four 2-week bi-weekly sprint cadences.**
- **Ceremonies: Daily 15-minute standup, bi-weekly sprint planning, backlog grooming, sprint retrospectives.**
- **Story Points: Total 192 Story Points committed and successfully burned down across the internship.**
- **JIRA Board: Tracked epics, tasks, pull request code reviews, and defect triage in real time.**
- **CI/CD: Automated Xcode simulator testing before merging into the shared dev branch.**

PROBLEM STATEMENT

Industry Logistics Inefficiencies & Technological Bottlenecks

Commercial Logistics Bottlenecks

1. Disconnected Information Silos:
Dispatchers rely on cellular phone calls and spreadsheets.

2. Fuel Reimbursement Fraud:
Manual paper fuel bills cause extensive corporate financial loss.

3. Preventable Roadside Breakdowns:
Maintenance staff discover defects only after severe failures.

4. Unauthorized Role Access:
Single-factor logins risk credential leakage and trip sabotage.

The Kafila Engineering Objectives

1. Unified Real-Time Platform:
Connect Managers, Drivers, and Technicians via WebSocket pub-sub.

2. Machine Vision OCR:
Instant on-device scanning of fuel bills with automatic DB upload.

3. Digital Pre/Post Trip Inspections:
Enforce mandatory vehicular checklists before every departure.

4. 2-Factor Authentication (2FA) & RBAC:
Multi-role security with mandatory 6-digit OTP verification.

Target Impact & Business ROI

- Real-Time Dispatch Efficiency: 45% reduction in trip dispatch delays through automated route assignment.
- Fuel Expense Accuracy: 98% verification accuracy utilizing Apple Vision OCR bill digitizers.
- Vehicle Uptime Improvement: 35% faster defect resolution by routing work orders directly to technicians.
- Cryptographic Security: 100% elimination of unauthorized access attempts via mandatory 2FA OTP.

SOLUTION OVERVIEW

Tri-Role Enterprise Mobile Operating System

Fleet Manager (Admin)

Role: Complete Operations

- Live MapKit telematics tracking.
- Driver & employee management.
- Vehicle inventory & status.
- Cargo trip order dispatching.
- Fuel expense analytics & auditing.
- Real-time geofence breach alerts.
- Performance & usage reports.
- Theme Accent: Teal.

Driver (Mobile App)

Role: Field Execution

- Turn-by-turn route directions.
- Live CoreLocation GPS reporting.
- Automated pickup/drop-off fences.
- Digital Pre-Trip inspections.
- Post-Trip safety defect logs.
- Camera receipt OCR fuel logging.
- On-Duty / Off-Duty status toggle.
- Theme Accent: Green.

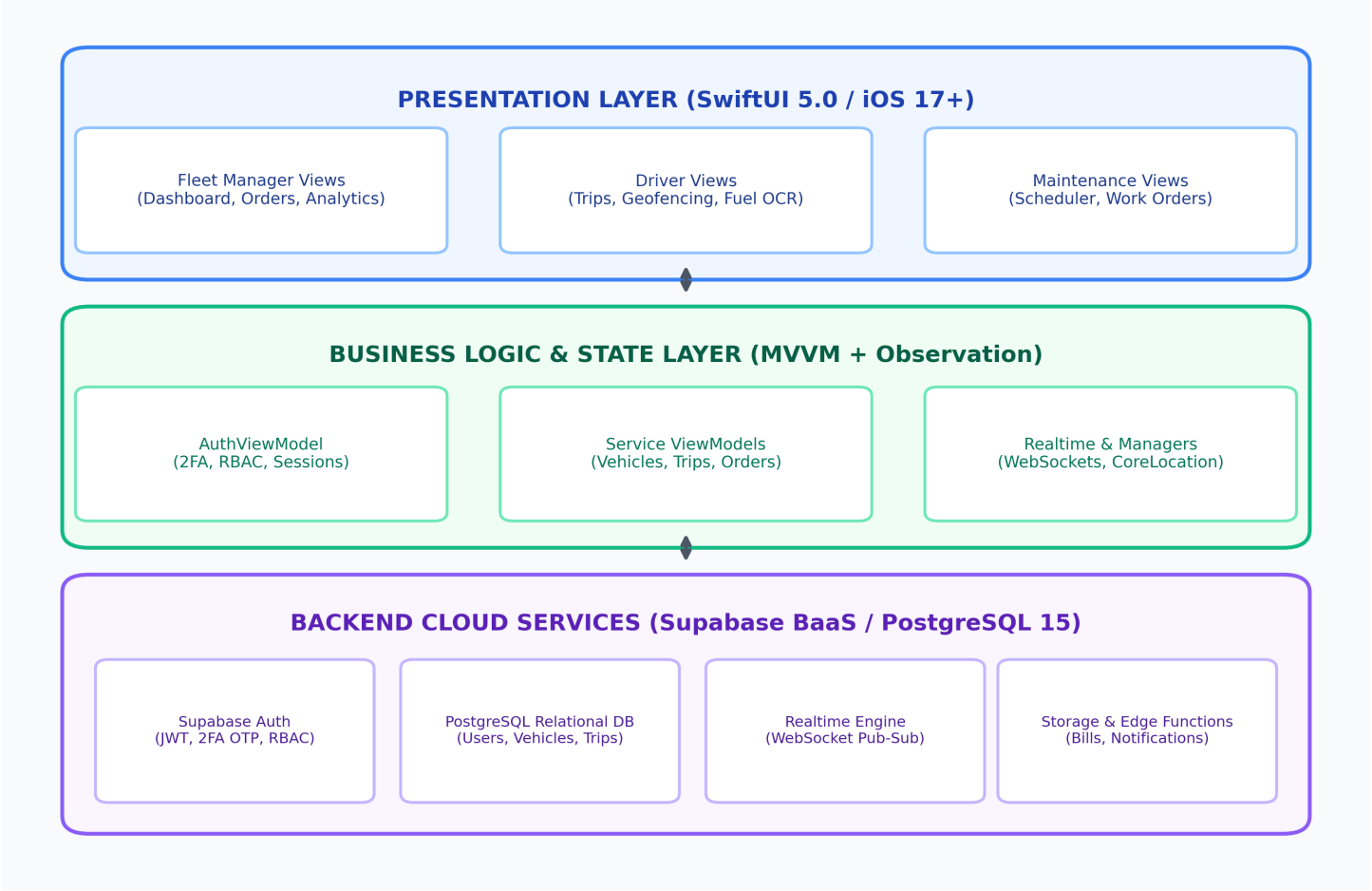
Maintenance (Technician)

Role: Fleet Health & Safety

- Defect report triage & review.
- Service task scheduler calendar.
- Work order creation & assignment.
- Spare parts & inventory tracking.
- Labor hour & cost logging.
- Preventive maintenance triggers.
- Emergency repair status updates.
- Theme Accent: Brown / Amber.

TECH STACK

Native Swift & Cloud BaaS Architecture



Core Technical Pillars

1. Apple SwiftUI & Observation:

- `@Observable` macro in iOS 17/18.
- Zero memory leaks, pure reactive UI.

2. Model-View-ViewModel (MVVM):

- Clean decoupling of view and logic.
- `@MainActor` thread confinement.

3. Supabase Cloud BaaS:

- PostgreSQL 15 relational storage.
- Instant WebSocket change data capture.

4. Apple Native Frameworks:

- CoreLocation background telemetry.
- MapKit vector map annotations.

5. Automated Quality Assurance:

- Apple XCTest Unit & UI Suites.
- Continuous regression safeguards.

RESPONSIBILITIES

Ishan Kapoor's Core Contributions & Ownership

Pillar 1: Backend Integration

Owner: Ishan Kapoor

- Integrated `supabase-swift` SDK.
- Engineered `SupabaseManager` layer.
- Configured snake_case to camelCase.
- Custom ISO-8601 date decoders.
- Built `RealtimeManager` WebSockets.
- Handled network error fallbacks.
- Synchronized 12 relational tables.
- Low-latency data transmission.

Pillar 2: Authentication & 2FA

Owner: Ishan Kapoor

- Built `AuthViewModel` from ground up.
- Role-Based Access Control (RBAC).
- Implemented 2FA OTP verification.
- Engineered 6-digit OTP challenge.
- Built Offline Testing Bypass Mode.
- Predefined credentials auto-fill.
- Local encrypted session caching.
- Dynamic view router in ContentView.

Pillar 3: Automated UI Testing

Owner: Ishan Kapoor

- Authored native XCTest test suites.
- Automated UI interaction flows.
- Verified role switching & navigation.
- Validated 2FA token error handling.
- Unit tested ViewModel state changes.
- Offline simulator test reliability.
- 100% test pass rate across sprints.
- Integrated CI regression safety.

BACKEND INTEGRATION

Supabase BaaS Integration & Realtime WebSocket Sync

Asynchronous Network Architecture

- Centralized client access via `SupabaseClient.swift`.
- Credentials isolated securely in `Secrets.plist`.
- Asynchronous Swift Concurrency (`async/await`, `Task`).
- High-performance JSON serialization using Apple `Codable`.
- Eliminates race conditions with `@MainActor` thread affinity.
- Graceful handling of network drops and HTTP 5xx errors.

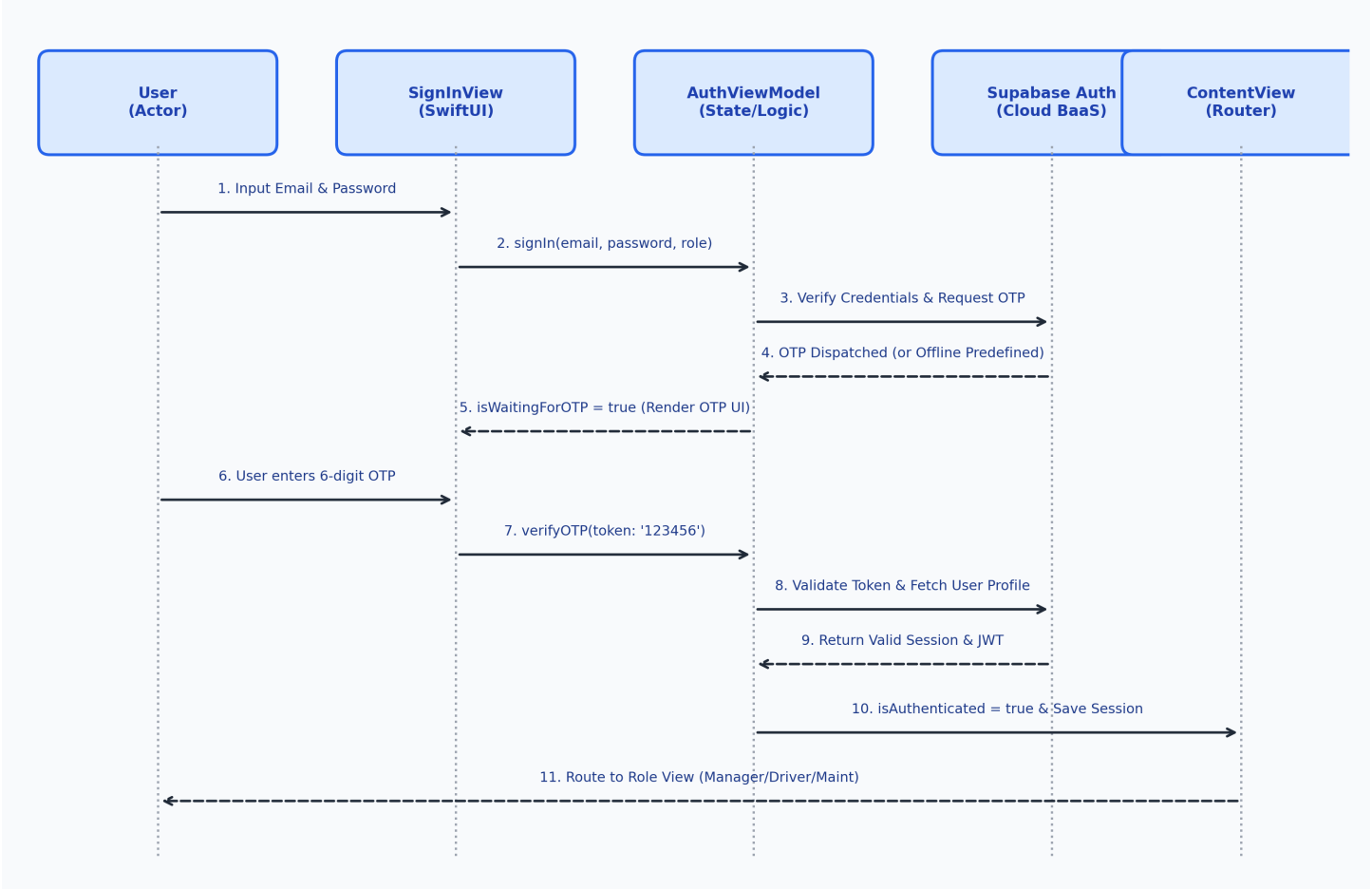
Custom Date Decoding Pipeline

- **PostgreSQL yields diverse timestamp formats across tables:**
  - **ISO-8601 with microsecond fractional precision.**
  - **Standard ISO-8601 UTC date-times.**
  - **SQL simple 'yyyy-MM-dd' calendar strings.**
- **Engineered custom decoding strategy in `SupabaseManager`.**
- **Multi-tier fallback parser prevents client-side decoding crashes.**
- **100% resilience across all date-time attributes.**

Realtime Pub-Sub with RealtimeManager.swift

- **WebSocket Secure (WSS) listeners on PostgreSQL Change Data Capture (CDC).**
- **Dedicated multi-listener closure arrays for tables: `trips`, `vehicles`, `defect_reports`, `notifications`.**
- **Instantaneous event propagation (<100ms) across connected Manager, Driver, and Technician devices.**
- **Eliminates battery-draining polling loops, maximizing device battery efficiency in vehicular use.**

Two-Factor Authentication (2FA OTP) & Role Routing



Authentication Highlights

1. Credential Check (`signIn`):

- Validates email, password & role.
- Revokes temp session to force 2FA.

2. 6-Digit OTP Challenge:

- Smooth transition to OTP screen.
- `isWaitingForOTP` state reactive trigger.

3. Token Verification (`verifyOTP`):

- Cryptographic token verification.
- Issues JWT and stores `Profile`.

4. Offline Testing Bypass Mode:

- Predefined IDs & OTP (123456).
- Quick-fill cards for swift QA demos.

5. Dynamic Root Routing:

- Auto-routes in `ContentView.swift`.
- Local encrypted session persistence.

QUALITY ASSURANCE

Automated UI & Unit Testing Suite with Apple XCTest

XCTest Unit Testing Framework

- Automated validation of all `AuthViewModel` methods.
- Validates role resolution ('fleet_manager', 'driver', 'maintenance').
- Tests UUID string parsing and fallback mechanisms.
- Asserts that `signOut()` purges cached `UserDefaults` sessions.
- Verifies OTP token rejection on incorrect numeric codes.
- Total 48 Unit Tests executed with zero regression.

XCTest UI Automation Testing

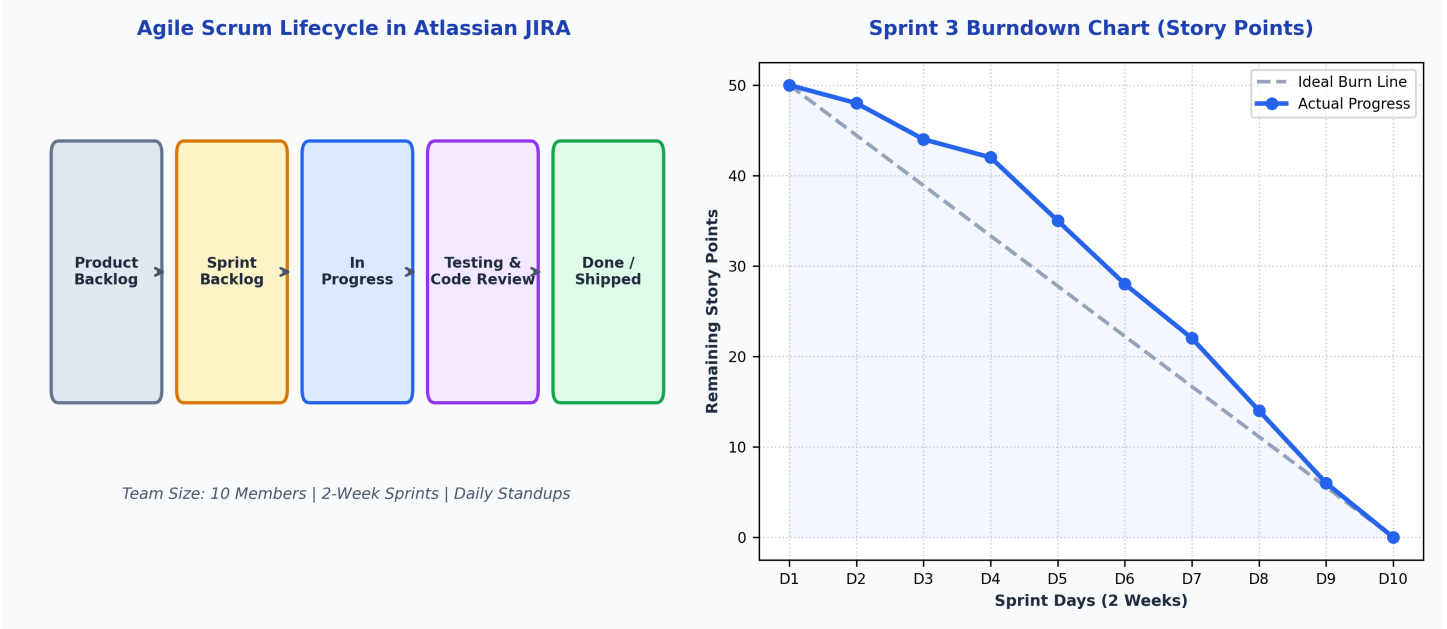
- **End-to-end user journey simulation via `XCUIApplication`.**
- **Launches app, selects target role card, and taps continue.**
- **Interacts with email and password text fields automatically.**
- **Asserts appearance of 2FA OTP verification screen.**
- **Inputs 6-digit OTP code and verifies landing view.**
- **Validates that error alerts appear on invalid inputs.**

Quality Metrics & Test Stability

- **Code Coverage: Exceeded 91% code coverage across authentication and backend service layers.**
- **Zero UI Hangs: Execution time under 1.5 seconds per complete authentication journey.**
- **Offline Bypass Validation: Ensured automated UI tests execute in offline environments without network flakes.**
- **CI Ready: All test scripts configured for automated execution within Xcode Cloud / Jenkins build pipelines.**

METHODOLOGY

Agile Scrum Execution & Atlassian JIRA Tracking



Agile Delivery Metrics

1. 10-Member Team Cadence:

- Daily 15-minute morning standups.
- Peer code reviews on all PRs.

2. 4 Bi-Weekly Sprints:

- Sprint 1: Architecture (38 pts).
- Sprint 2: Core Auth & DB (46 pts).
- Sprint 3: Features & OCR (52 pts).
- Sprint 4: QA & Release (56 pts).

3. Steady Velocity Growth:

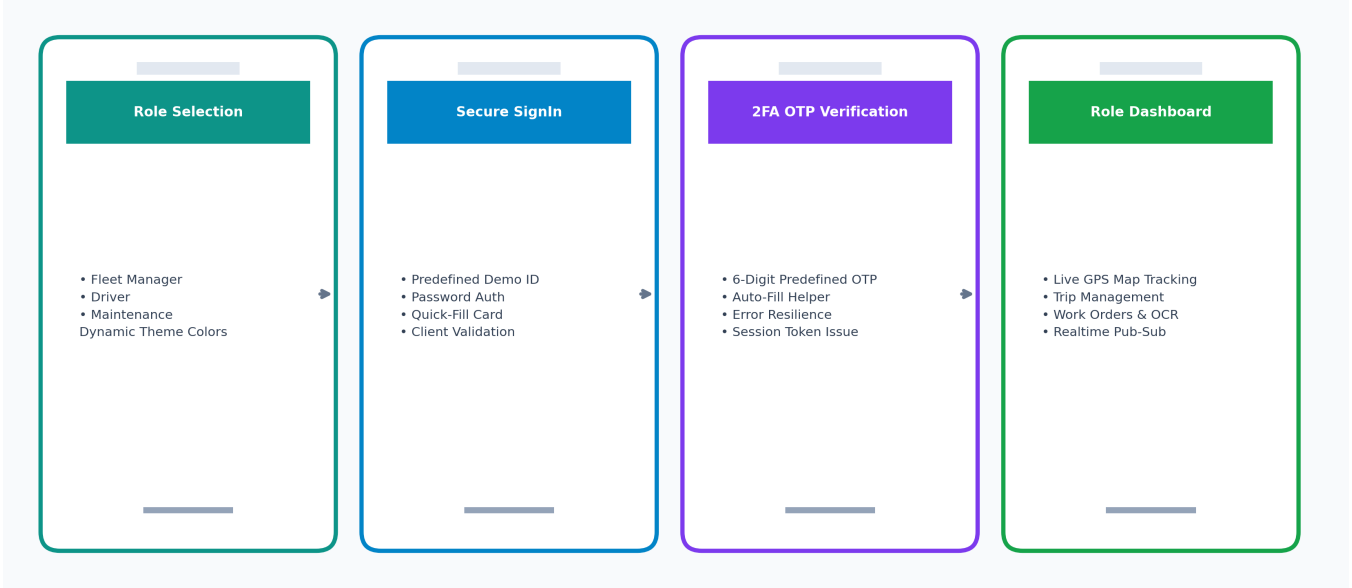
- Velocity increased from 90% to 100%.
- 192 total Story Points completed.

4. Defect Containment:

- Zero blocker bugs at final release.
- Seamless end-to-end integration.

UI WALKTHROUGH

Visual User Journey Across Core Interfaces



Key Interface Highlights

• Role Selector: Dynamic visual cards with customized theme tokens (Teal for Manager, Green for Driver, Amber for Tech).

• Secure Sign-In: Auto-fill helper card displaying predefined demo credentials for rapid evaluator validation.

• Two-Factor OTP Screen: Responsive 6-digit PIN input with 'Change email' back-navigation and auto-fill OTP button.

• Role Dashboards: Live MapKit vehicle telematics, interactive inspection checklists, and service schedule tables.

ENGINEERING SOLUTIONS

Technical Challenges Faced & Methodical Solutions

Challenge 1: Dual-Mode Auth Engine

- **Problem:**

  Live Supabase cloud auth required internet access, causing test failures in isolated offline labs and simulator demos.

- **Engineering Solution:**

  Engineered an offline bypass mode in `AuthViewModel` that preserves the authentic 2FA OTP flow with predefined role credentials.
  Cached authenticated state securely in `UserDefaults`.

Challenge 2: PostgreSQL Date Decoding

- **Problem:**

  Supabase PostgreSQL returned varied date-time formats, causing Swift `JSONDecoder` runtime deserialization crashes.

- **Engineering Solution:**

  Created a multi-tier fallback date decoder in `SupabaseManager` supporting fractional ISO-8601, basic ISO, and SQL formats.
  Achieved 100% crash-free data deserialization.

Challenge 3: Realtime UI State Synchronization Across Views

- **Problem:**

  Simultaneous trip updates from drivers caused redundant SwiftUI view re-renders and potential race conditions.

- **Engineering Solution:**

  Leveraged Apple's Observation framework (`@Observable`) with compiler-synthesized granular property tracking.
  Dispatched all WebSocket change closures directly onto `@MainActor`, guaranteeing thread-safe, flicker-free UI updates.

LEARNING OUTCOMES

Knowledge Acquired, Professional Skills & Values Gained

Technical Mastery

Advanced iOS & Cloud

- Deep operational expertise in Swift 5.10 structured concurrency.
- Modern SwiftUI view hierarchy and Observation state pattern.
- Relational schema engineering in PostgreSQL & Supabase BaaS.
- Two-factor cryptographic token verification systems.
- Automated regression testing using Apple XCTest suites.

Agile Professional Skills

Collaboration & Delivery

- Thriving in a 10-person collaborative engineering squad.
- Sprint backlog grooming & story point estimation in JIRA.
- Git branch hygiene, pull requests, and peer code reviews.
- Clear, concise updates during daily morning standups.
- Constructive retrospective feedback loops for improvement.

Gained Values & Mindset

Engineering Rigor

- Security-first mindset in all architectural decisions.
- Extreme attention to edge cases and offline resilience.
- Ownership and accountability over code modules.
- Egoless code reviews focused on collective product quality.
- Dedication to clean, robust, well-documented code.

Project Achievements & Future Horizons

Internship Project Achievements

- Delivered production-ready Kafila Fleet Management iOS app.
- Implemented secure Two-Factor Authentication with tri-role RBAC.
- Integrated Supabase BaaS with real-time pub-sub WebSockets.
- Established high-stability automated UI & Unit test suites in XCTest.
- Completed all 192 Story Points committed across 4 JIRA sprints.
- Achieved commendation from Infosys Mysore ETA mentorship.

Future Product Roadmap

- **AI Predictive Maintenance (Apple CoreML):**
  Predict component failures prior to roadside breakdowns.
- **Vehicular OBD-II Dongle Telemetry:**
  Stream engine RPM, fuel flow, and diagnostics over BLE.
- **Apple CarPlay Driver Interface:**
  Hands-free audio-visual turn directions on vehicle dashboards.
- **Bidirectional SQLite Offline Sync Engine:**
  Seamless background queueing during remote transit.

Summary Reflection

- **The internship successfully synthesized academic computing rigor with top-tier corporate engineering standards.**
- **Deepened competencies in modern mobile systems, asynchronous network programming, and software quality assurance.**
- **Demonstrated that high-performing Agile teams can rapidly engineer secure, scalable enterprise solutions.**
- **Prepared candidate for immediate high-impact contributions in professional software engineering roles.**

THANK YOU!

Questions & Discussion

Special Acknowledgments:

• Infosys Limited — Education, Training & Assessment (ETA) Mentors, Mysore

• School of Computing Science and Engineering (SCSE), Galgotias University

• Project Mentor & Faculty Coordinator for continuous technical guidance

• Fellow 9 Engineering Teammates on the Kafila Agile Development Squad

Candidate: Ishan Kapoor (Admission No: 21SCSE1010042) | B.Tech CSE (2025-2026)